Module 4 Data Exploration and Preparation

7.1 Module Overview

Module Number	4	**Module Title**	Data Exploration and Preparation					
Stage of Principal Programme			Award	**Semester**	1 PT 1 FT	**Duration.** ***(Weeks F/T)***	13	**ECTS**	10
Mandatory / Elective (M/E)	M	**Hours of Learner Effort / Week**[6]	5 PT 7.5 FT					

Analysis of required hours of learning effort		
Teaching and Learning Modalities	**✓if relevant to this module**	**Approx. proportion of total (hours)**
Contact Hours		
On-site face-to-face	✓	6 PT 81 FT
Synchronous online	✓	48 PT 0 FT
Synchronous Hybrid	X	0
Indirect/Non-contact Hours		
Asynchronous	✓	6 PT 9 FT
Work Based	X	0
Other (Identify)	✓	190 PT 160 FT
Total		250
Pre-Requisite Module, if any. Module Number and Title	None	

[6] Average (over the duration of the module) of the contact hours per week i.e. effort while in contact with staff

Co-Requisite Module, if any. Module Number and Title	None
Maximum number of learners per instance of the module	60

Specification of the qualifications (academic, pedagogical and professional/occupational) and experience required of staff working in this module.

Role e.g., Tutor, Mentor, Lecturer, Research Supervisor, etc.	**Qualifications & experience required**	**Staff (X) : Learner (Y) Ratio Express as X:Y**
Lecturer	Minimum NFQ Level 9 cognate award is normally required. However NFQ Level 8 in a cognate award is acceptable in cases where significant industrial experience is evident.	1:60

Assessment Techniques – percentage contribution

Continuous Assessment	100%	**Proctored Exam – in person**	0	**Practical Skills Based**	0
Project	0	**Proctored Exam - online**	0	**Work Based**	0
Capstone modules (Yes/No)?	No	**If Yes, provide details**			

7.2 Minimum Intended Module Learning Outcomes (MIMLOs)

MIMLO On successful completion of this module a learner will be able to:	Related MIPLO #
1. Gather and format data into an actionable resource dependent upon the tools and resources available ensuring compliance with ethical, legal and regulatory standards	3, 5, 6, 7, 8
2. Investigate the data using Exploratory Data Analysis (EDA) and discuss the findings with a view to how this will affect decision making.	1, 2, 5
3. Implement a data cleaning and preparation plan using appropriate techniques for further analysis	2, 3, 4
4. Select and employ appropriate data engineering processes to enhance insight and scope, of the data analysis	2, 4, 5
5. Communicate the properties of the data, ensuring consistency and replicability, (Qualitative, Quantitative, Data Type, Feature Type, etc.) using a Data Dictionary	2, 3, 4, 7
6. Demonstrate a project cycle, using tools that ensure evidence of verifiable ongoing and regular progress, and incorporating feedback from stakeholders.	5, 6, 7, 8

7.3 Indicative Module Content, Organisation and Structure

Week	Topic	Anticipated TLA Activity
1	**Introduction (to include installation and overview of required digital tools)** **What is Data?** **Understanding Data** ● Variables/Observations ● Attributes **Data Management** ● Primary/Secondary ● What/Where/How of Data Acquisition ● Version Control	Acquisition (Lecture) Practice (Install Tools) Collaboration (Group Task)
2	**Data Management** ● Consistency and Documentation (Data Dictionary)	Discussion (Review of previous weeks work) Acquisition (Lecture)

Week	Topic	Anticipated TLA Activity
	● Legal, Ethical and Regulatory constraints **Exploratory Data Analysis (EDA):** ● Visualisation ● Statistical Properties ● Univariate/Multivariate	Investigation (Research how Legal, Ethical and Regulatory constraints differ geographically and are influenced by the problem domain)
3	**Exploratory Data Analysis (EDA)** ● What is the data telling us and how do we use that information? **Planning our Preparation** ● CRISP-DM ● SEMMA ● Version Control for validation and Project Management	Discussion (Review of previous weeks work) Acquisition (Lecture, Demonstration) Practice (Practical implementation of EDA/Version Control)
4	**Data Cleaning** ● Documenting your process ● Missing Data ○ Dropping ○ Imputation ○ Flagging	Discussion (Review of content to date) Acquisition (Lecture, Demonstration) Production (Review CA Progress)
5	**Data Cleaning** ● Feature Selection ● Duplicates ● Outliers	Discussion (Review of content to date) Acquisition (Lecture, Demonstration) Practice (Practical implementation of Data Cleaning)
6	**Data Cleaning** ● Malformed data ○ Numerical ○ Categorical ○ Text Based	Discussion (Review of previous weeks work) Acquisition (Lecture, Demonstration) Practice (Practical implementation of Data Cleaning) Production (Review CA Progress)
7	**READING WEEK**	**Acquisition**
8	**Feature Engineering** ● Discretization ○ Grouping of equal intervals	Discussion (Review of previous weeks work) Acquisition (Lecture, Demonstration)

Week	Topic	Anticipated TLA Activity
	○ Grouping based on equal frequencies (of observations in the bin) ○ Grouping based on decision tree sorting (to establish a relationship with target)	Practice (Practical implementation)
9	**Feature Engineering** ● Categorical Encoding ○ Count and Frequency encoding ○ Ordinal encoding ○ One Hot Encoding	Discussion (Review of previous weeks work) Acquisition (Lecture, Demonstration) Practice (Practical implementation) Discussion (Class feedback on CA Submission)
10	**Feature Engineering** ● Scaling ○ Min-Max Scaling ○ L2-Normalisation ○ Standardisation/Variance Scaling	Discussion (Review of content to date) Production (Review CA Progress) Acquisition (Lecture, Demonstration) Practice (Practical implementation)
11	**Feature Engineering** ● What is the Curse Of Dimensionality? ● Dimensionality Reduction ○ PCA ○ LDA	Discussion (Review of previous weeks work) Acquisition (Lecture, Demonstration) Practice (Practical implementation)
12	**Feature Engineering** ● Feature Creation/Extraction ○ Aggregates ○ Mathematical Operations ○ Deriving new features from existing ones	Discussion (Review of previous weeks work) Acquisition (Lecture, Demonstration) Practice (Practical implementation) Production (Review CA Progress)
13	**Review of all Content to date** **Open Q&A**	Discussion (Review of content to date)

7.4 Work based learning and practice-placement (if applicable)

Not applicable.

7.5 Specific module resources required (if applicable)

There are no module-specific resources required beyond the standard resources required for other elements of the programme (e.g., laptop, classrooms, and virtual learning environment).

7.6 Application of programme teaching, learning and assessment strategies to this module

The TLA strategy for this module includes:

- Fully utilising a blended (hybrid) learning environment, ensuring students benefit from a blend of structured synchronous and asynchronous learning activities;
- Use of ABC Learning Design model to create a more inclusive and diverse learning experience across all six learning activity types;
- An integrated and authentic continuous assessment strategy that is scaffolded vertically and horizontally with interactive feedback loops to inform learning;
- Allowing formal and informal feedback mechanisms to inform the TLA in an agile and flexible way throughout the academic year; and
- Informed by CCT College's Centre for Teaching and Learning (CTL) recommendations, initiatives and events.

For specific TLA strategies across all learning types in the ABC learning design model see 7.3 column 2 above. This is indicative, and will be flexible enough to respond to the learners requirements. In addition the asynchronous delivery scheduled will comprise of dedicated material, activities and contributions that students will be directed to engage with throughout the trimester.

7.7 Summative Assessment Strategy for this module

Assessment Type	Assessment Description	MIMLO addressed	MIPLO addressed	% weighting of total grade	Week of assessment submission
Continuous Assessment	CA1: EDA and Planning Practical Artefact, 1000 Word Report	1, 2, 5, 6	1, 2, 3, 4, 5, 6, 7, 8	40%	7
Continuous Assessment	CA2: Cleaning, engineering, and Validation Building on CA 1 Practical Artefact, 1500 Word Report	3, 4, 5, 6	1, 2, 3, 4, 5, 6, 7, 8	60%	13

7.8 Sample Assessment Materials

A representative sample of each assessment type for modules whose grade contributes to the overall grade classification is provided in Appendix H. Sample Assessment. These sample assessments are accompanied by criterion referenced grading schemes illustrating how learner's responses to assessment tasks will be graded.

7.9 Indicative reading lists and other information resources

Core Resources:

- Ball, R. and Rague, B. (2023) The Beginner's Guide to Data Science. 1st ed. 2022 edition. Springer.
- Foxwell, H.J. (2020) Creating Good Data: A Guide to Dataset Structure and Data Representation. 1st ed. edition. New York: Apress.

- McGregor, S. (2022) Practical Python Data Wrangling and Data Quality: Getting Started with Reading, Cleaning, and Analyzing Data. 1st edition. Sebastopol, CA: O'Reilly Media.
- Narayana, D., Ranjan, S. and Tyagi, N. (2023) Basic Computational Techniques for Data Analysis. 2nd edition. London New York: Routledge India.
- Publishing, A.I. (2020) Data Preprocessing with Python for Absolute Beginners: Step-by-Step Guide with Hands-on Projects and Exercises. AI Publishing LLC.

Supplementary Resources:

- ACM SIGs foster international technical communities within the major areas of computing. (no date). Available at: https://www.acm.org/special-interest-groups (Accessed: 29 August 2022).
- Bruce, P.C. and Bruce, A. (2017) Practical statistics for data scientists: 50 essential concepts. First edition. Sebastopol, CA: O'Reilly.
- Godsey, B. (2017) Think like a data scientist: tackle the data science process step-by-step. Shelter Island: Manning.
- Heatley, L.M. (no date) National Centre for Text Mining — NaCTEM. NaCTeM. Available at: http://www.nactem.ac.uk/software.php (Accessed: 29 August 2022).
- Ilyas, I.F. and Chu, X. (2019) Data cleaning. First edition. New York, NY: Association for Computing Machinery (ACM Books, 28).
- Jonge, E. de and Loo, M. van der (2018) Statistical data cleaning with applications in R. Hoboken, NJ: John Wiley & Sons.
- Matthes, E. (2019) Python crash course: a hands-on, project-based introduction to programming. 2nd edition. San Francisco, CA: No Starch Press.
- McCallum, Q.E. (2013) Bad data handbook: mapping the world of data problems. 1. ed. Beijing, Köln: O'Reilly.
- Squire, M. (2015) Clean Data. Birmingham, UK: Packt Publishing (Community Experience Distilled). Available at: https://search.ebscohost.com/login.aspx?direct=true&db=nlebk&AN=996715&site=eds-live (Accessed: 29 August 2022).